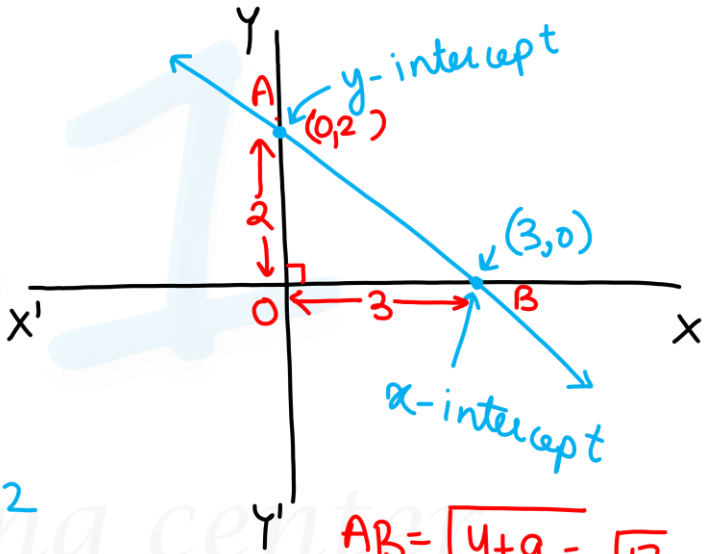
Intercept: (अंत:खंड/प्रतिछेद/अवरोधन)

Eg:

$2x+3y=6$

x intercept, $y=0 \Rightarrow x=3$

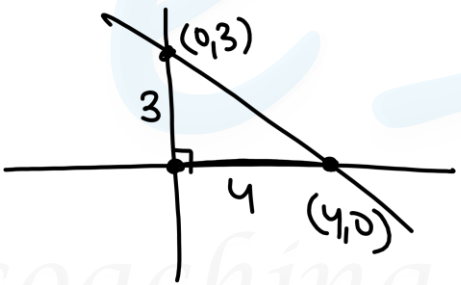
y intercept, $x=0, \Rightarrow y=2$



coaching centre

x intercept, $y=0 \Rightarrow x=4$

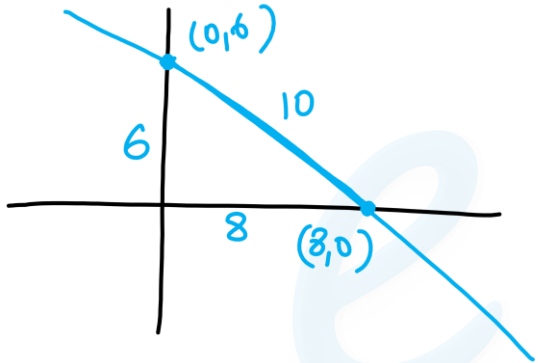
y intercept, $x=0 \Rightarrow y=3$



1. Find the intercepts made by the line $3x + 4y - 12 = 0$ on the axes :
रेखा $3x + 4y - 12 = 0$ द्वारा अक्सों पर बनाये गये प्रतिछेदों को ज्ञात करो?

a) 2 and 3
b) 4 and 3
c) 3 and 5
d) NOT

coaching center



2. The shortest length of the intercept between the x-axis and y-axis of the equation $6x + 8y - 48 = 0$ is:

समीकरण $6x + 8y - 48 = 0$ के $x-$ अक्ष और $y-$ अक्ष के बीच अंतःखंड की सबसे छोटी लंबाई कितनी है?

a) 10 units
b) 6 units
c) 14 units
d) 12 units

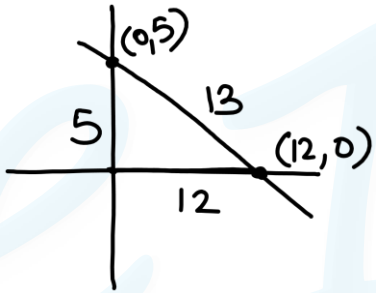
coaching center

3. The length of the portion of the straight line $5x + 12y = 60$ intercepted between the axes is:

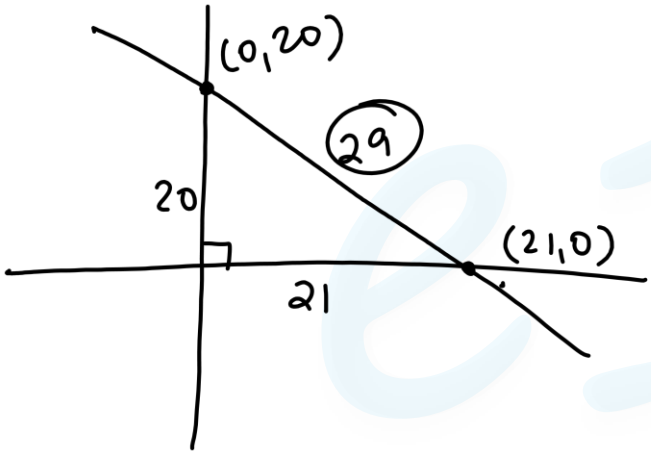
सरल रेखा $5x + 12y = 60$ के अक्षों के बीच अंतःखंडित (intercepted) भाग की लंबाई ज्ञात कीजिए।

a) 5 b) 13

c) 17 d) 12



coaching center



4. If the graph of the equation $20x + 21y = 420$ cuts the coordinate axes at P and Q, then what is the measure of the length (in units) of PQ?

यदि समीकरण $20x + 21y = 420$ का ग्राफ निर्देशांक अक्षों को P और Q पर काटता है, तो PQ की लंबाई की माप (इकाइयों में) क्या है?

a) 25 b) 29

c) 40 d) 35

coaching center

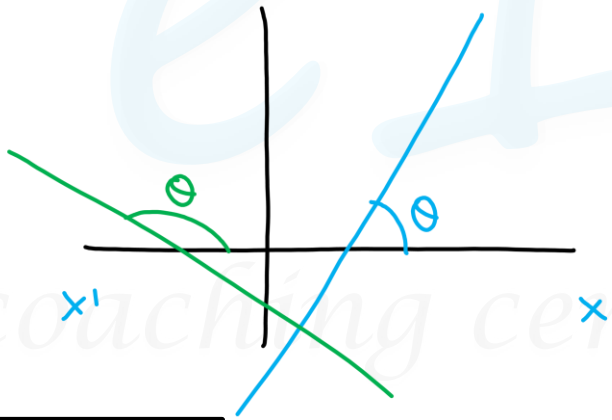
Slope of a line
(रेखा का ढलान)

coaching center

Slope/gradient of a line: (ढाल/ढलान/प्रवणता)

- **Slope of a line represents its degree of inclination.**

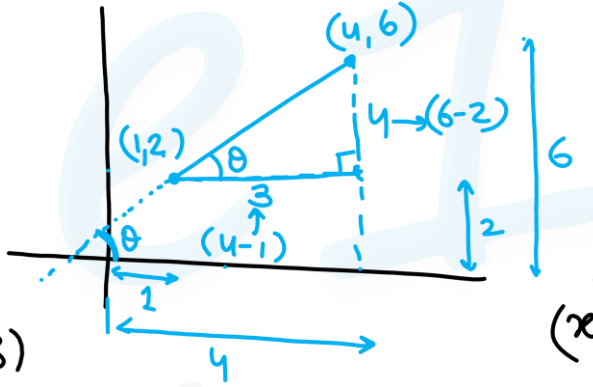
→ angle of line with +ve direction of x-axis.



coaching center

i) slope = $m = \tan\theta$

ii)



$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{y_1 - y_2}{x_1 - x_2}$$

(x_2, y_2)

(x_1, y_1)

(2, 3) (5, -3)

$\frac{6}{-3} = -2$

coaching center

iii) $ax + by + c = 0$

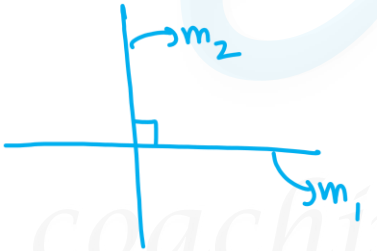
$$\text{Slope} = -\frac{a}{b} = \frac{-(\text{Coeff. of } x)}{\text{Coeff. of } y}$$

Eg: $4x + 7y - 18 = 0$

$$m = \frac{-4}{7}$$

coaching center

- **Slope of $x-axis$ is 0.**
- **Slope of $y-axis$ is not defined.**
- **Slopes of two parallel lines are same $i.e. m_1 = m_2$**
- **Slopes of two perpendicular lines are negative reciprocals of each other $i.e. m_1 = -\frac{1}{m_2}$ or $m_1 \times m_2 = -1$**



$m_1 = \frac{2}{3}$ $m_2 = \frac{-3}{2}$

$m_1 = \frac{-7}{8}$ $m_2 = \frac{8}{7}$

coaching center

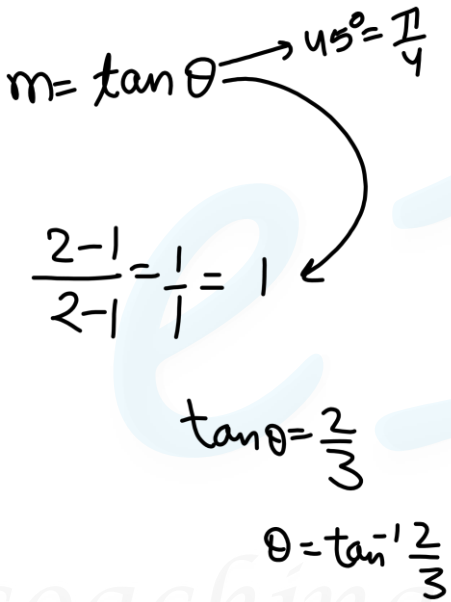
$\frac{7-5}{9-7} = \frac{2}{2} = 1$

5. Find the slope of the line joining the points $(7, 5)$ and $(9, 7)$:
$(7, 5)$ और $(9, 7)$ के मिलाने से बनी रेखा का ढलान ज्ञात करो|

a) 1 b) 2

c) $\frac{1}{2}$ d) 3

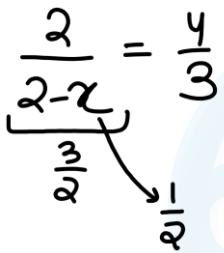
coaching center



6. What is the slope of line joining the points $(1, 1)$ and $(2, 2)$?
$(1, 1)$ और $(2, 2)$ के मिलाने से बनी रेखा का ढलान ज्ञात करो|

a) $\frac{\pi}{2}$
b) $\frac{\pi}{4}$
c) $\frac{\pi}{6}$
d) $\frac{\pi}{8}$

coaching center



7. Slope of the line AB is $4/3$. Co-ordinates of points A and B are $(x, -5)$ and $(2, -3)$ respectively. What is the value of x?

रेखा AB की ढलान $4/3$ है। बिन्दुओं A और B के निर्देशांक क्रमशः $(x, -5)$ और $(2, -3)$ है । x का मान क्या है

a) $1/2$ b) $-1/4$

c) 4 d) -4

coaching center

8. What is the slope of the line $5x + 3y = 12$

रेखा $5x + 3y = 12$ का ढाल क्या है?

a) $2/5$

b) $3/5$

c) $-3/5$

d) $-5/3$

$-\frac{5}{3}$

e1 coaching center

$$+\frac{a}{3} = +\frac{2}{3}$$

9. $ax + 3y = 6$ has slope $-2/3$ What is the value of a?

$ax + 3y = 6$ का ढलान $-2/3$ है। a का मान क्या है?

a) -2 b) 2

c) 3 d) -3

e1 coaching center

$$\frac{3-(-3)}{6-2} = \frac{6}{4} = \frac{3}{2}$$

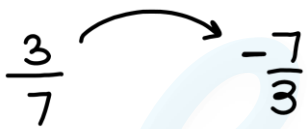
10. A line passes through the points A $(2, -3)$ and B $(6, 3)$. Find the slope of the line which is parallel to AB ?

बिन्दुओं A $(2, -3)$ और B $(6, 3)$ से एक रेखा गुजर रही है। तो उस रेखा के समांतर रेखा की ढलान क्या है?

a) $-2/3$ b) $3/2$

c) $1/2$ d) $3/4$

coaching center



11. What is the slope of the line perpendicular to the line passing through the points $(3, 5)$ and $(-4, 2)$?
उस रेखा का ढाल बताइए जो बिन्दुओं $(3, 5)$ और $(-4, 2)$ से गुजरती हुई रेखा के लम्बवत है?

a) $-3/5$ b) $-2/3$
c) $-7/3$ d) $7/3$

coaching center

$m_1 = \frac{-2}{-5} = \frac{2}{5}$

$m_2 = -\frac{5}{2}$

12. What is the slope of line perpendicular to the line $2x - 5y = 12$

रेखा $2x - 5y = 12$ के लम्बवत रेखा की ढलान क्या है?

a) $2/5$ b) $5/2$

c) $-2/5$ d) $-5/2$

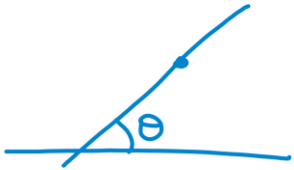
e1 coaching center

Equation of a line
(रेखा की समीकरण)

coaching center

Different ways to form equation of a line:

- Slope point form
- Two-point form
- Intercept form
- Slope intercept form





e1 coaching center

Slope point form:

i) Ex: $(2,3)$ $m = -\frac{4}{3}$

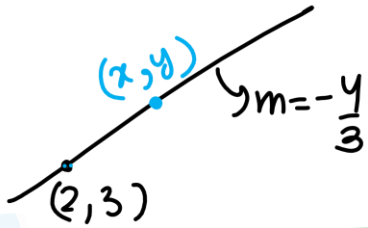
value (variable)

$\text{slope} = \text{slope}$

$$-\frac{4}{3} = \frac{y-3}{x-2}$$

$$\Rightarrow -4x+8 = 3y-9$$

$$\Rightarrow 4x+3y-17=0$$



$ax+by+c=0$

slope

$4x+3y=17$

8+9

coaching center

ii) $(4_1, -2)$ $m = -\frac{1}{2}$ (x, y)

$$-\frac{1}{2} = \frac{y+2}{x-4}$$

$$\Rightarrow -x+4 = 2y+4$$

$$\Rightarrow x+2y = 0$$

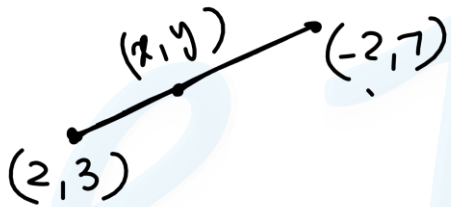
$x+2y = 0$

$4-4=0$

coaching center

Two-point form:

i) Ex



$$m = \frac{7-3}{-2-2}$$

$$m = \frac{4}{-4} = \frac{-1}{1}$$

$x+y=5$

Value = Variable (x,y)

$$\text{Slope} = \text{Slope}$$

$$-1 = \frac{4}{-4} = \frac{y-3}{x-2}$$

$$\Rightarrow -x+2 = y-3$$

$$\Rightarrow x+y-5=0$$

ii)



$m = \frac{-1}{5}$

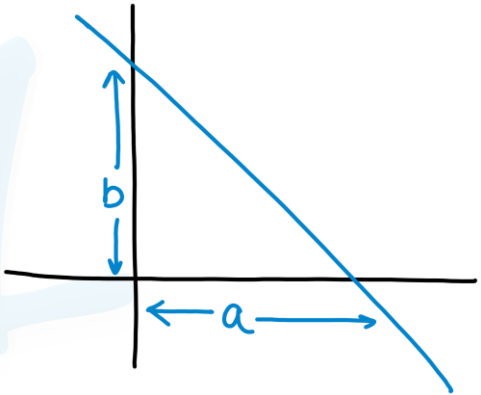
$x + 5y = -17$

3-20

coaching center

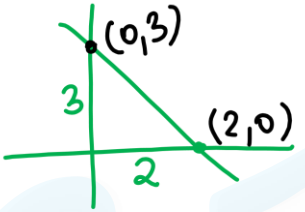
Intercept form:

$$\frac{x}{a} + \frac{y}{b} = 1$$



e1 coaching center

i)



$$\frac{x}{2}+\frac{y}{3}=1 \quad \Big\} \times 6$$

$$3x+2y=6$$

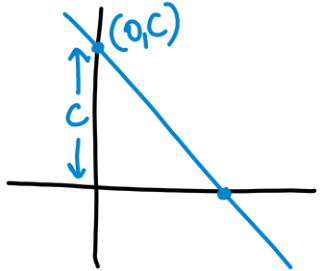
e1 coaching center

Slope intercept form:

m

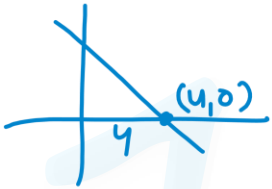
y intercept $= c$

$$y = mx + c$$



coaching center

i) Ex $m = \frac{7}{3}$



$-7x+3y=-28$

$-28+0$

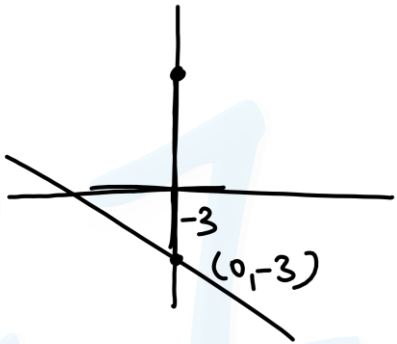
$7x-3y=28$

coaching center

ii) $m = \frac{5}{3}$

$y = mx + c$

$y = \frac{5}{3}x - 3$



$-5x + 3y = -9$

$0 - 9$

coaching center

$$m = \frac{4}{-4} = \frac{-1}{1}$$

$$x + y = -3$$

$$-1 - 2 = -3$$

13. Find the equation of the line through the points $(-1, -2)$ and $(-5, 2)$:
बिन्दुओं $(-1, -2)$ और $(-5, 2)$ से गुजरने वाली रेखा का समीकरण होगाः

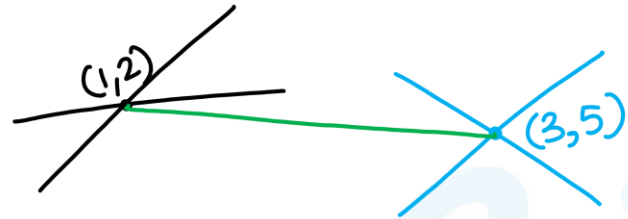
a) $2x + y = 3$

b) $3x - 2y + 7 = 0$

c) $x + y + 3 = 0$

d) NOT

coaching center



$3x = 3$

$x = 1$

$y = 2$

$3x = 9/3$

$y = 5$

$m = \frac{5-2}{3-1} = \frac{3}{2}$

$-3x + 2y = 1$

$-3 + 4 = 1$

14. Find the equation of the line joining the points of intersection of $2x + y = 4$ and $x - y + 1 = 0$ with point of intersection of $2x - y - 1 = 0$ and $x + y - 8 = 0$:
जो रेखा $2x + y = 4$ और $x - y + 1 = 0$ के प्रतिछेदक बिंदु को $2x - y - 1 = 0$ और $x + y - 8 = 0$ के प्रतिछेदक बिंदु से मिलाती है उसका समीकरण क्या होगा।

a) $2x + 3y - 6 = 0$

b) $3x + 2y - 1 = 0$

c) $3x - 2y + 1 = 0$

d) None of these

coaching center

$-5x+4y=-22$

$-10-12$

15. Find the equation of the line passing through the point $(2,-3)$ and having its slope $5/4$:
बिंदु $(2,-3)$ से गुजरती हुई और $5/4$ झुकाव वाली रेखा का समीकरण बताओ।
a) $4x-5y=23$
b) $10x-8y=5$
c) $5x-4y=22$
d) NOT

coaching center

$m = \tan 60 = \frac{\sqrt{3}}{1}$

$-\sqrt{3}x + y = -3\sqrt{3} - 4$

$-3\sqrt{3} - 4$

16. Find the equation of the line which passes through the point $(3, -4)$ and makes an angle of $60°$ with the positive direction of $x - axis$:

उस सरल रेखा का समीकरण ज्ञात कीजिये जो बिंदु $(3, -4)$ से गुजरती है और $x -$ अक्ष की धनात्मक दिशा के साथ $60°$ का कोण बनाती है।

a) $3x - \sqrt{3}y = 4$

b) $\sqrt{3}x - y = 4 + 3\sqrt{3}$

c) $x\sqrt{3} + y = 3\sqrt{2} + 5$

d) NOT center

e1 coaching centre

$m_1 = \frac{6-2}{5-1} = \frac{4}{4} = 1$

$m_2 = -\frac{1}{1}$

$x+y = -9$

$-4-5$

17. Find the equation of the line passing through the point $(-4, -5)$ and perpendicular to the line joining the points $(1, 2)$ and $(5, 6)$:

बिंदु $(-4, -5)$ से गुजरने वाली उस रेखा का समीकरण बताइए जो की बिन्दुओं $(1, 2)$ और $(5, 6)$ को मिलाने से बनने वाली रेखा के लम्बवत है?

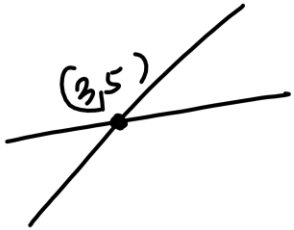
a) $x + y + 17 = 0$

b) $3x + 2y + 11 = 0$

c) $x + y - 2 = 0$

d) NOT

coaching center



$5x = 15$

3

$y = 5$

$m_1 = \frac{6-4}{5-2} = \frac{2}{3}$

$m_2 = -\frac{3}{2}$

$3x + 2y = 19$

$9 + 10 = 19$

18. Find the equation of the straight line which passes through the point of intersection of the straight lines $x + y = 8$ and $3x - 2y + 1 = 0$ and is perpendicular to the straight line joining the points $(2, 4)$ and $(5, 6)$:

उस रेखा का समीकरण ज्ञात करो जो की बिन्दुओं $(2, 4)$ और $(5, 6)$ के मिलाने से बनी रेखा के लम्बवत है और रेखाओं $x + y = 8$ और $3x - 2y + 1 = 0$ के प्रतिछेदक बिंदु से गुजरती है।

a) $3x - 2y - 19 = 0$

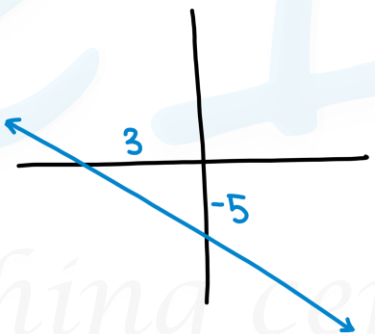
b) $3x + 2y = 19$

c) $6x - 4y + 8 = 0$

d) NOT

$$\frac{x}{3} + \frac{y}{-5} = 1 \quad \Big\} \times 15$$

$$5x - 3y = 15$$



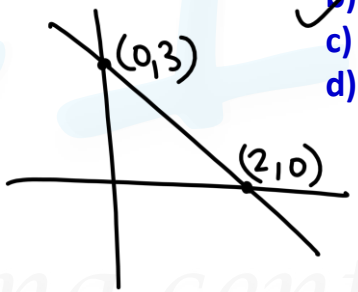
19. What is the equation of a line which has 3 as x –intercept and -5 as y –intercept?
एक रेखा के समीकरण का पता लगायें, जिसकी x – प्रतिच्छेद 3 और y प्रतिच्छेद 5 है?

a) $3x - 5y = 15$
b) $5x - 3y = 15$
c) $5x + 3y = 15$
d) $3x + 5y = 15$

coaching center

$$\frac{x}{2} + \frac{y}{3} = 1 \quad \Big\} \times 6$$

$$3x + 2y = 6$$



20. Find the equation of the line which cuts off intercepts 2 and 3 from the axes:
अक्षों पर 2 और 3 प्रतिच्छेद बनाने वाली रेखा का समीकरण बताओ।

a) $9x - 7y = 6$
b) $3x + 2y = 6$
c) $4x + 3y = 7$
d) NOT

coaching center